



Introduction to “Artemis” HPC

The University of Sydney’s High Performance Computer, called *Artemis*, is a 4264 core Infiniband-connected cluster accessible to all researchers at the University of Sydney. Our system aims to achieve the following:

- Job wait times should be roughly proportional to requested walltimes
- Support the most promising research projects that require HPC, in line with the University’s strategic goals¹
- Host compute resources for faculties/schools who would otherwise buy and maintain their own hardware

When Artemis was first launched in 2015, it had 1464 cores available in one resource pool with 3 month maximum walltimes, and two 24 core high memory nodes, running a very diverse workload from many users. This resulted in:

- High utilisation (above 90%)
- Long wait times for jobs of all sizes

The long wait times were a pain point for our users, so we redesigned Artemis’s structure in November 2016 in an attempt to reduce job wait times, particularly for short-running jobs. To achieve this, we divided Artemis’s compute resources into three main resource pools: fair share, strategic allocations, and condominiums, as shown in Table 1 below. We also applied per-user limits on cores based on walltime requests in “fair share” to stop one user monopolising resources, shown in Figure 1.

Table 1. A summary of the three main compute resource pools on Artemis.

Fair Share	Strategic Allocations	Condominiums
• All Artemis users may submit jobs here • Subdivided into “small”, “normal”, “large”, “small-express”, and specialty (high memory, GPU, and interactive) resource pools • Dividing the fair share pool into small, normal and large should make small jobs start quickly, but long jobs may have to wait • Uses the Fair Share² algorithm of PBS Professional to assign job priority	• Dedicated compute resources granted to winners of the University’s “HPC Grand Challenge Scheme” • Allocated to outstanding researchers to help them perform at their full potential, in line with the University’s strategic direction¹ • Current allocations are solving problems in computational chemistry, chemical engineering, transport logistics, gene and stem cell research, and fluid dynamics	• Dedicated compute resources for groups who financially contribute to Artemis • Relieves faculties/schools of the burden of managing their own compute resources • Artemis currently has one condominium

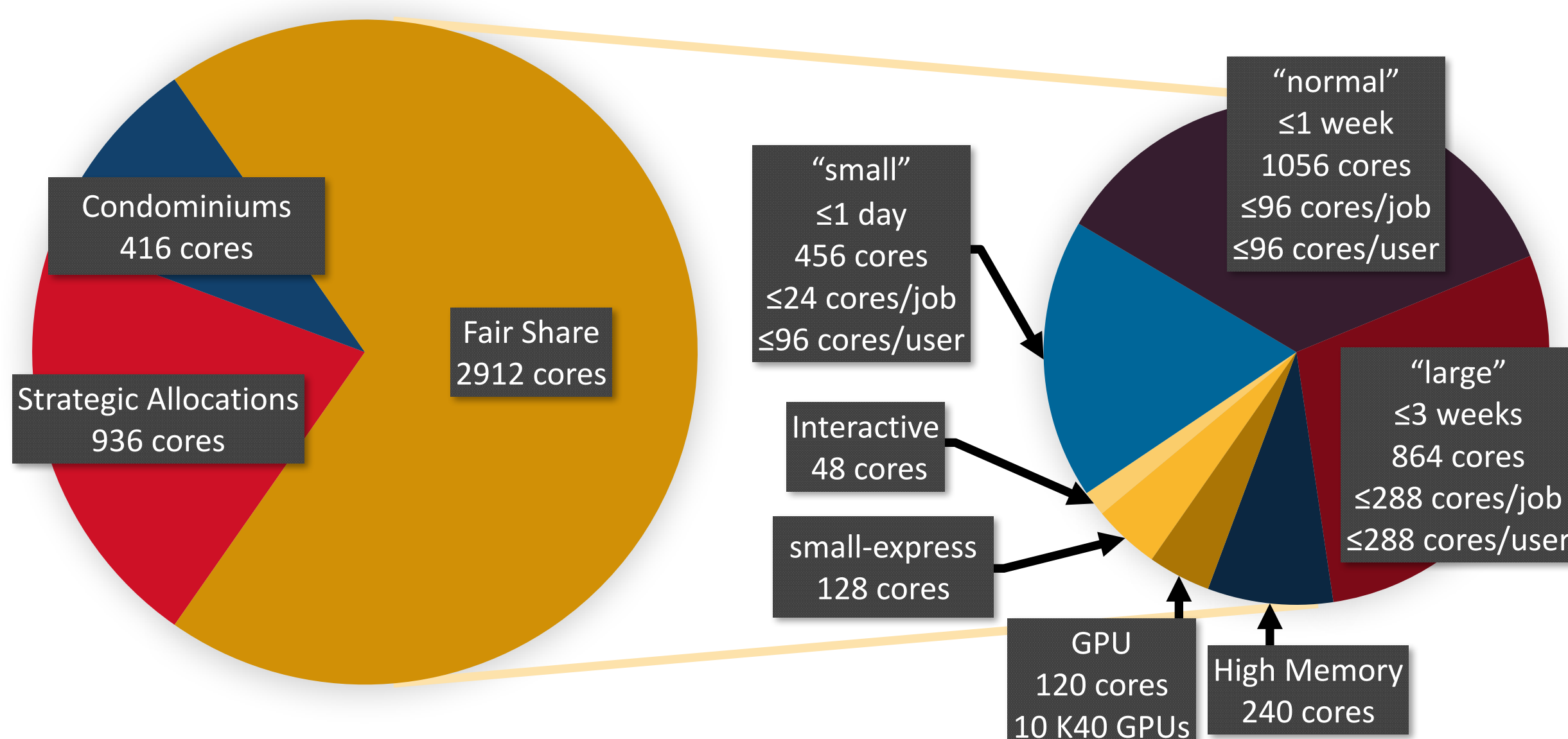


Figure 1. A breakdown of Artemis’s compute pools. The pie chart on the right shows how the “Fair Share” resources are subdivided.

Artemis Job Scheduling

- Separation of jobs into multiple resource pools can reduce overall system utilisation
- The impact of resource segregation is reduced by introducing “Peering” and the “Scavenger” queue

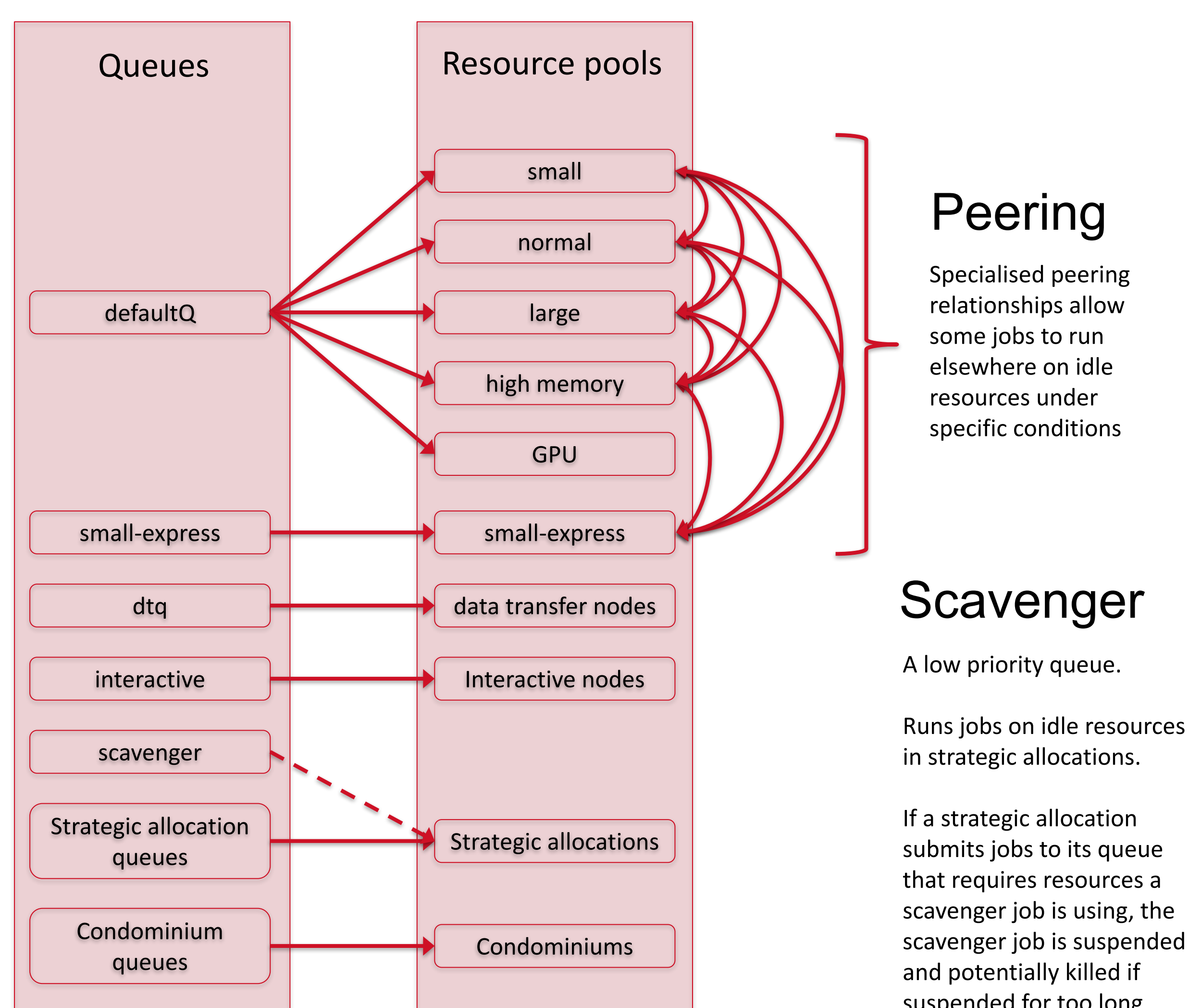


Figure 2. Diagrammatic representation of how Artemis allocates jobs from submission queues to the underlying resource pools.

System utilisation and wait times

In the figures below, the n^{th} percentile wait time means $n\%$ of jobs started in less than y hours. Wait times are only one of the metrics we use to estimate user experience. We are focussing on this metric in this poster as it was a key pain point when Artemis first launched. We have not solved the “wait time” problem, but we hope the data below provides insight into our experience with this queue structure.

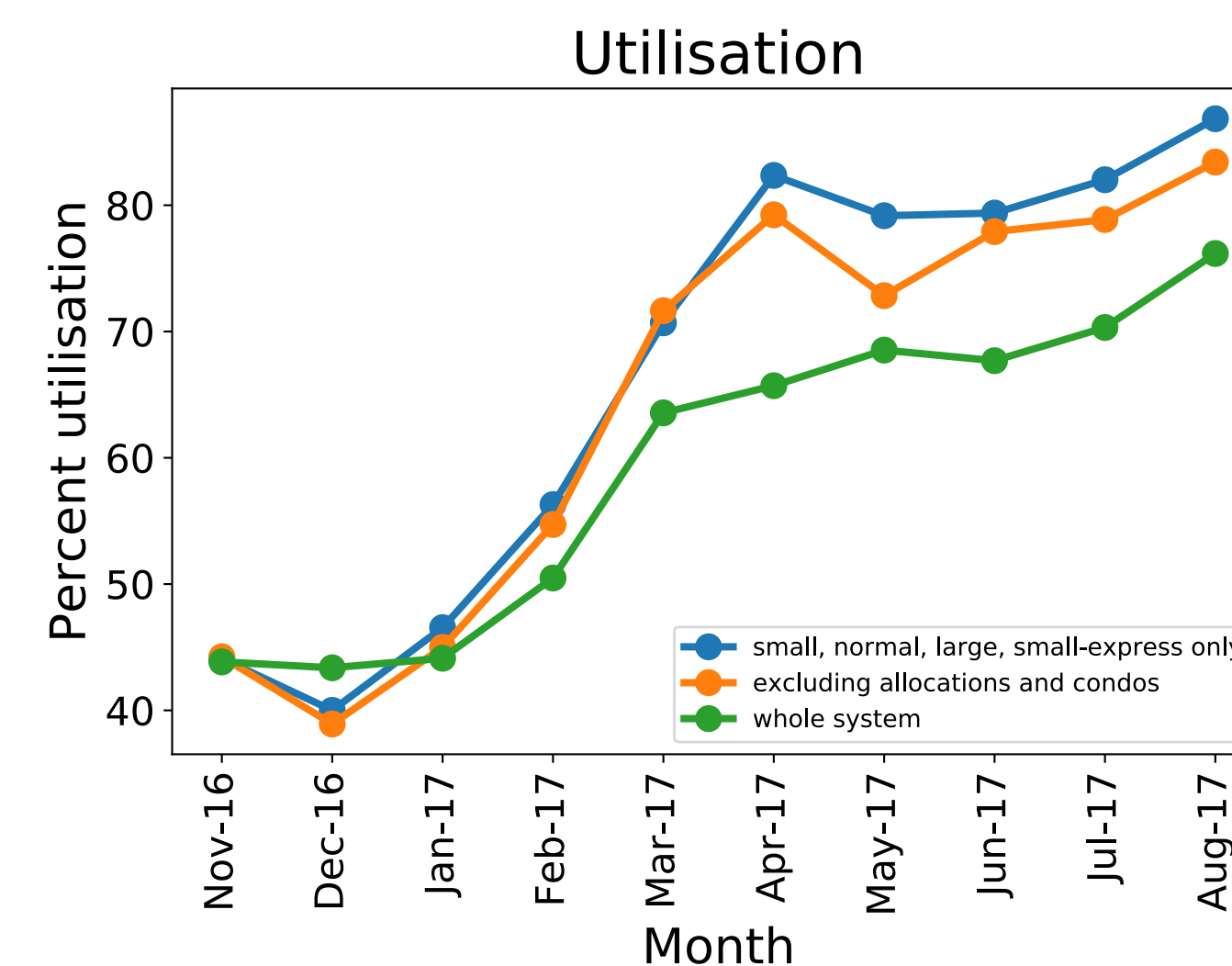


Figure 3. System utilisation for parts of Artemis shown in the legend in the figure. Lines are guides for the eye only.

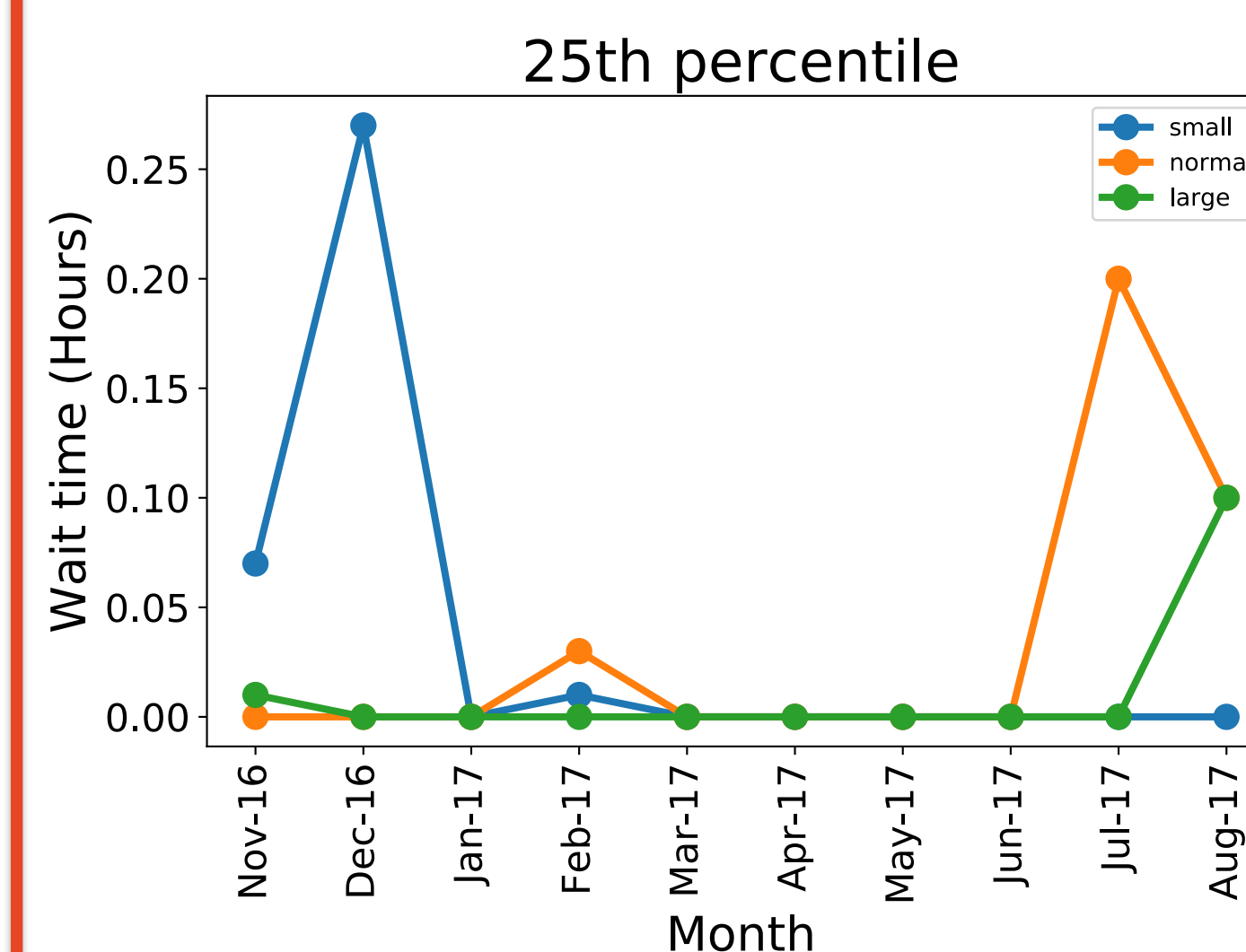


Figure 4. 25th percentile job wait times for small, normal, and large job classifications. Lines are guides for the eye only.

- 25% of jobs have started in less than an hour for small, normal and large job classifications..
- 25th percentile wait times are marginally above zero for July and August, but are not yet a cause for concern.

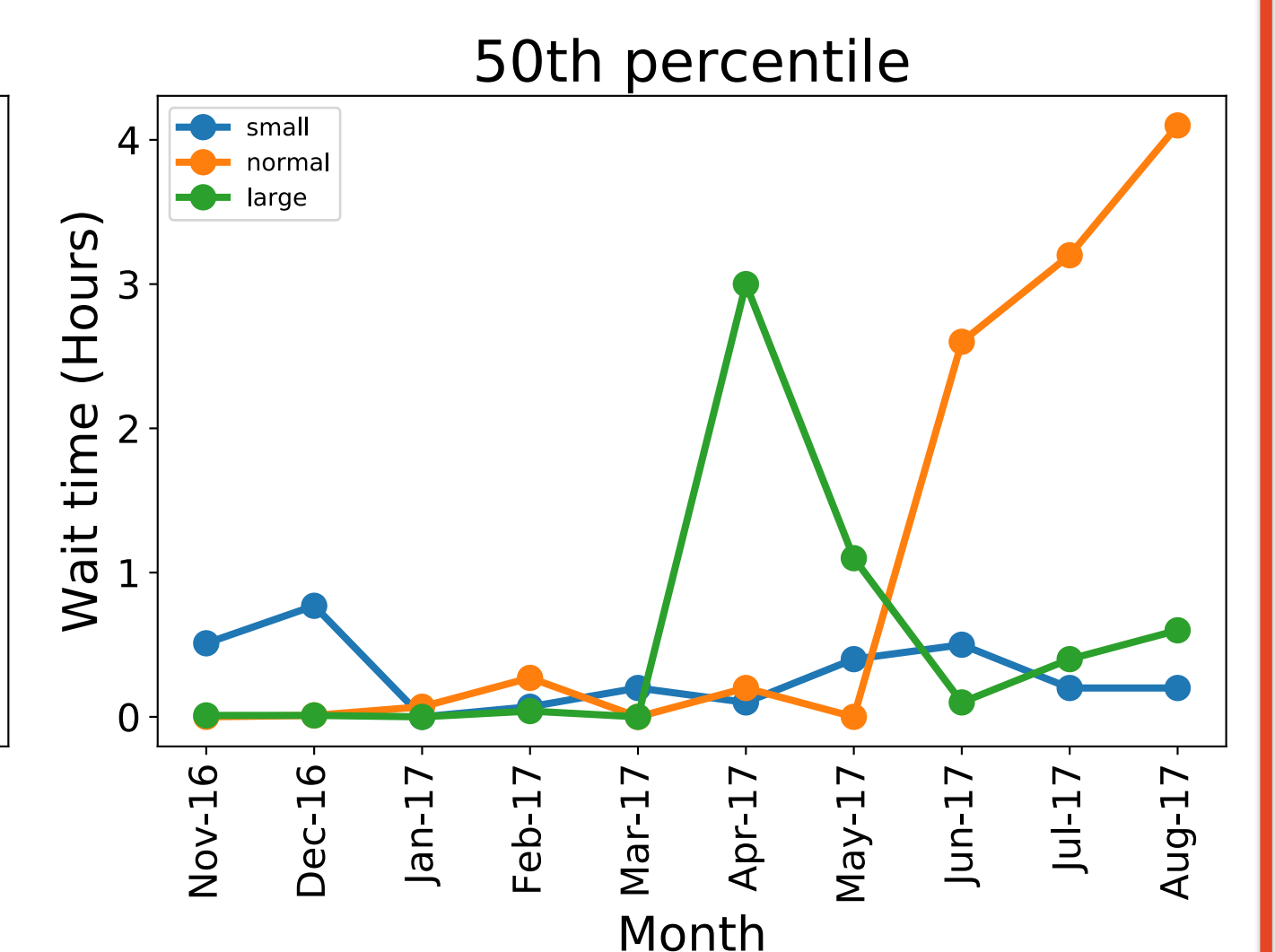


Figure 5. 50th percentile job wait times for small, normal, and large job classifications. Lines are guides for the eye only.

- 1 hour wait times in small in November and December were likely caused by per-user limits preventing all queued jobs starting, not due to high utilisation.
- Half of all jobs on Artemis have started in less than 5 hours.
- Normal wait times are increasing, while small and large are relatively low.

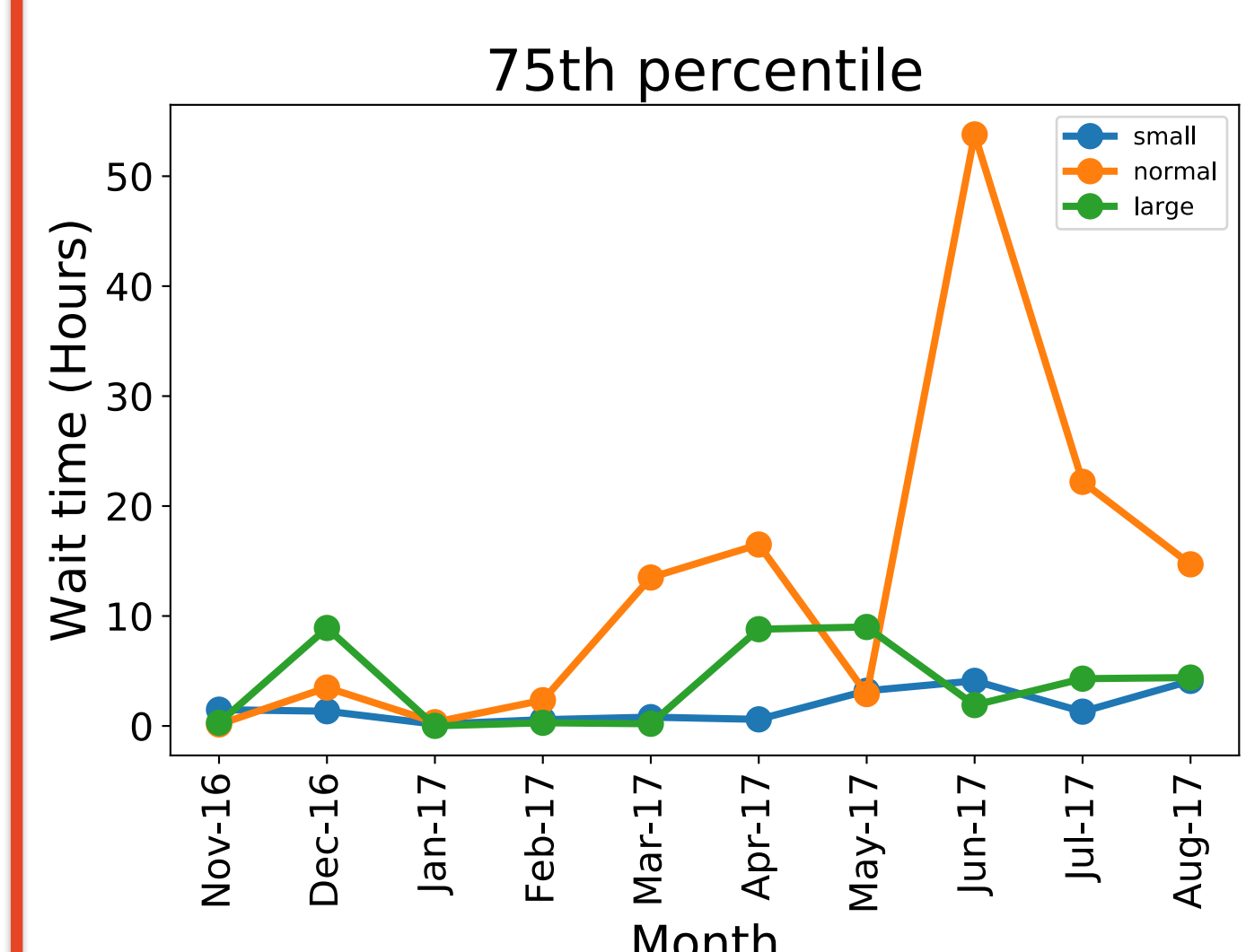


Figure 6. 75th percentile job wait times for small, normal, and large job classifications. Lines are guides for the eye only.

- 75% of jobs each month have started in less than 25 hours, except for in June where 75% of jobs in normal started in less than 54 hours.
- Small and large wait times were low (less than 5 hours, respectively) indicating good throughput for 75% of these jobs.
- The normal wait time spike in June is indicative of strong demand for “normal” sized jobs.

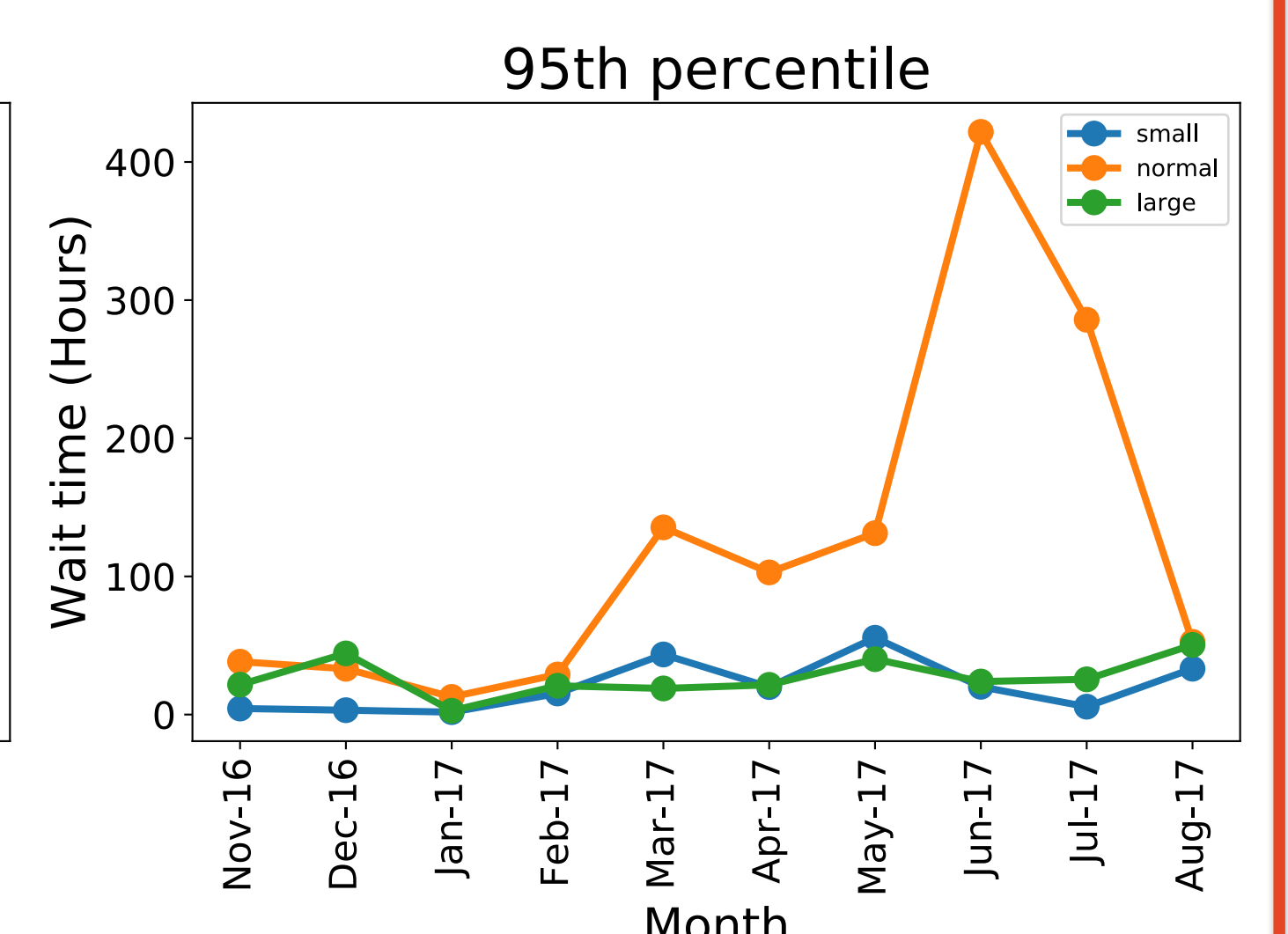


Figure 7. 95th percentile job wait times for small, normal, and large job classifications. Lines are guides for the eye only.

- 95% of jobs in small and large are starting in less than 50 hours. This is better than expected since “fair share” can result in very long queue times for heavy users.
- Some jobs in normal have waited more than 400 hours in June! This reduced to less than 50 hours by August, indicating there was high demand for “normal” jobs in June.

- Dedicated support to outstanding research, at the cost of some CPU utilisation, mitigated by peering and the scavenger queue.
- Wait times less than 24 hours for 75% of jobs each month, except in June 2017. Some jobs are still waiting a long time.

Acknowledgements

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